A Method to Select a Good Setting for the kNN Algorithm when Using it for Breast Cancer Prognosis

Alberto Palacios Pawlovsky[1] and Mai Nagahashi[2]

***Abstract*— Breast cancer is the world's second most frequent type of cancer and in Japan it is the third most frequent one. The prognosis of its recurrence, after a first treatment, is very important to increase the survival rate of a patient. This work shows the application of the k-Nearest Neighbors (kNN) method to prognosis breast cancer and also proposes a method to select a good setting with the parameters that can be changed when using this classification method. Using our method with the Wisconsin's breast cancer prognosis data, the kNN method has an average accuracy of 76%, a small standard deviation, and a small difference between its maximum and minimum values.**

***Index Terms*— prognosis tool, breast cancer, machine learning, classification method, kNN.**

I. Introduction

Breast cancer is the most frequent type of cancer among women in Japan and every year almost 50,000 women have this type of cancer and its mortality rate raises to nearly 24%. After a first treatment, it is very important to follow the development of it and detect if cancer will recur to take additional measures to eliminate it or stops its spreading. Therefore, prognosis is very important to increase the possibility of surviving of a patient.

There are studies that use clinical data to forecast the time of recurrence. One research work frequently cited in this field is the method based on linear programming of Mangasarian et al. [1]. Its authors developed a program to estimate the time to recurrence of breast cancer and their results are usually used as a base line. The data used in that paper is now known as the Wisconsin Breast Cancer Data and is available in three types from the UCI repository [2].

There are also studies to determine not the time to recurrence, but if there is going to be a recurrence or not [3]∼[6], and others that try to predict cancer survival [7]. These research works use machine learning and data mining techniques for the processing of several types of cancer related data [8]∼[15]. In the case of prediction, some works show results with an accuracy higher than 90% [15].

The kNN algorithm is a classification method frequently used in machine learning because it is very simple to implement and can be used in many ways [14]. We have been studying and using the kNN algorithm and found that the range of its results is very ample and prone to vary according to the settings of some of its parameters.

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In this paper we show some details when using the kNN algorithm for prognosis using the UCI's Wisconsin's breast cancer prognosis data. We processed this data in three different ways and used them to develop one method of selecting a good set of parameters (k, classification set size, and number of runs) for the algorithm.

In the following section we briefly explain the kNN algorithm, its parameters, and three possible ways of preprocessing data. In section III we detail our method to select one good setting for the algorithm. In the last section we show some conclusions and topics for further research.

II. Breast Cancer Prognosis using kNN

The k-Nearest Neighbor (kNN) algorithm is an unsupervised learning algorithm used for classification. It uses part of the target data for classification and the rest of it as the test set. This algorithm measures the distance (or similarity) of a new datum to the data for classification and chooses the k-closest ones. Then, it finds the class that has majority in that group and assigns it to the new datum. In Figure 1 we

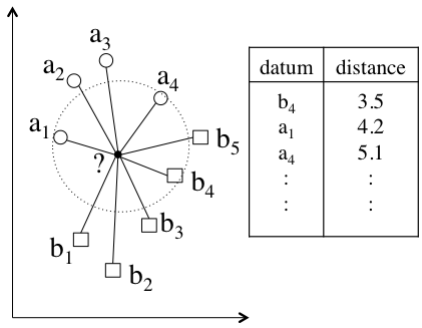

Fig. 1. Example of the use of the kNN algorithm.

show a 3−NN small example where the new datum (black dot) will be classified either as an a type or b type data. In this example the (Euclidean) distance of the new datum to all other data is calculated and sorted to find the three closest ones (see table in the figure). Two of them are of type a and only one of type b, so the new datum will be classified as a type a datum (distance-weighted voting).

We have implemented the kNN algorithm allowing us to change the percentage of the data used in the classification set, the number of neighbors (k), the type of preprocessing applied to the data, and the number of simulation runs with a given setting.

Data could be used without any preprocessing (raw data), could be standardized or normalized. For standardization, we

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subtract the mean and divide by the standard deviation each numeric value in the data [14]. For normalization we apply the following formula to each raw value.

$$x_{norm} = \frac{x_{raw} - x_{min}}{x_{max} - x_{min}} \quad (1)$$

This formula scales all the numeric data to the range $[0, 1]$.

The measure of similarity or distance can be implemented in many ways. The authors of [15] have studied different distances and classification rules for the kNN algorithm and found that the best results are obtained with the Euclidean and Manhattan distances. We use the Euclidean distance in our implementation.

The classification data is randomly chosen. So, for different classification sets with the same percentage of all the data, we will get different results. Therefore, for characterizing a classification set of a given size, we have to repeat its running several times and measure its average, maximum and minimum prediction accuracy results. Figure 2 shows the average results of running the Wisconsin breast cancer prognosis data with values of k ranging from 1 to 10, and nine classification set sizes. Even values of k show lower

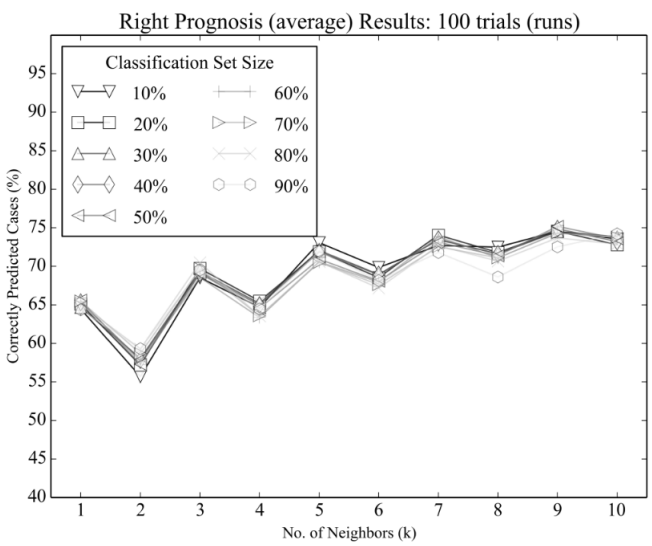

Fig. 2. Results with raw data and 10 classification set sizes : 100 runs.

average results because, in a draw, the algorithm chooses the first category in the sorted list of distances. We run simulations with classification data sets having sizes from 10% to 90% of all data. In the Wisconsin breast cancer prognosis data there are 198 cases of which 47 are recurrent cases. It has 4 cases with missing values (the number of affected lymph nodes). Some researchers replace those values by their mean but we opted for excluding them [12].

We can also see that average values increase with the number k of neighbors. Figure 3 shows the results with 1000 runs. The average values, compared to those of 100 runs, change very slightly and are imperceptible. Almost all research papers using kNN or other algorithms publish only the best or average values obtained with those algorithms. The results of figures 2 and 3 indicate that a higher value

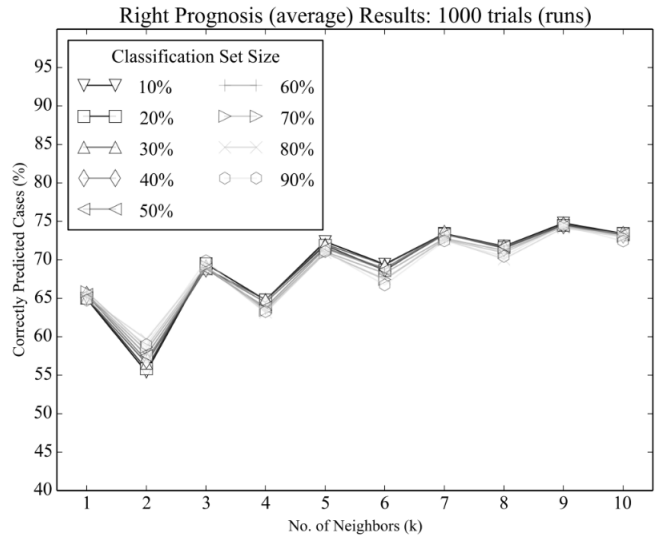

Fig. 3. Results with raw data and 10 classification set sizes : 1000 runs.

of k gives better average accuracy results, and for values beyond 10 we can expect some improvement, but very small. However, they tell us nothing about the minimum or maximum accuracy values.

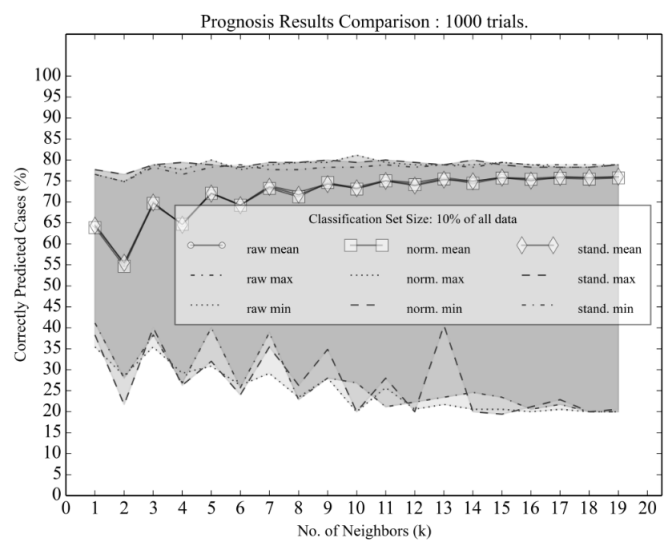

Fig. 4. Results with raw and preprocessed data: 10% set size.

Figure 4 shows the maximum, minimum, and avergae accuracy results of simulation runs using raw data, standardized data and normalized data for a training set using 10% of the available data. The black solid line confirms that the number of correctly predicted cases (average values) reach a peak near 73% for a 19-NN implementation, but also that this algorithm could be right in only one of five cases.

Figure 5 shows the results obtained with a classification set that uses 40% of the data. The average values do not change much, but we could see that the differences between maximum and minimum, the range of values, are smaller for almost all k values.

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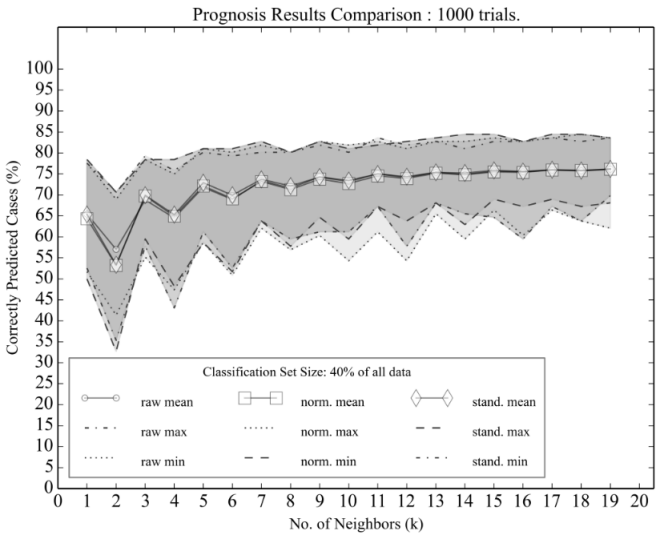

Fig. 5. Results with raw and preprocessed data: 40% set size.

However, if we further increase the percentage of data used in the classification set the differences between maximum and minimum values will again be large.

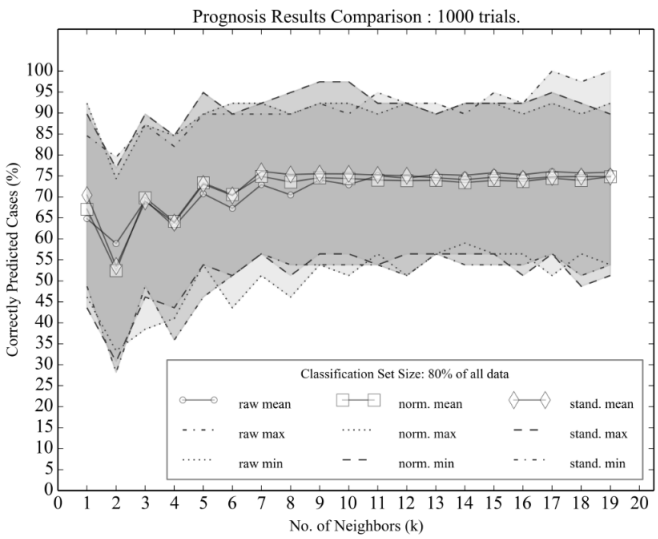

Fig. 6. Results with raw and preprocessed data: 80% set size.

Figure 6 shows these changes for a classification set that uses 80% of the data. The variations on the number of cases correctly predicted (average values), specially for small values of k and different data preprocessing, are a little bit clearer in this setting. This figure also shows us that we can expect higher maximum values, with some of them even going up to the 100% (17 and 19 k value settings).

Figures 4 to 6 also show us that there exists a very little difference in using the data in raw, normalized or standardized form. So, according to these results we can say that preprocessing does not play a key role neither influences much the results of the kNN algorithm. What we would like to have is a setting that gives us a high average value, and has also high minimum and maximum values. The selection of an optimal or near optimal (good) setting is hard to do from the output obtained, so we have developed a method to select one. This method is detailed in the following section.

III. Method to select a good setting for kNN.

The number of possible combinations of k and sizes of the classification set is too large to manually select a setting for prognosis of new data. Therefore, we have developed a method that, based on the results obtained with the test sets, selects a good setting. Before detailing it, we briefly explain the characteristics of the Wisconsin data for prognosis.

It has records of 198 patients and each record has 35 values. The first value is the identification number of the patient, the second one is a class value indicating the recurrence or not of cancer ('R': recurrent, 'N' : non recurrent). The third value shows the time to recurrence or the disease free time in months. The next 30 real values are computed from a breast mass sample taken from the patient. The first 10 of them are mean values of features of the nuclei of the cells in the sample. The next 10 ones are standard error values of these features and the last 10 are the maximum values of them. The last 34th and 35th values in a record show the tumor size and the number of affected lymph nodes, respectively.

We used all the above last 32 features and the class value for classification. We excluded four records missing the lymph nodes values. Since the smallest training set size is of 10% of all the data, the maximum value of k is limited to 19 (maximum number of neighbors).

TABLE I

Items, scores and weight values for setting's selection.

Item	score	weight
k	1:1, 3:2, ..., 17:9, 19:10	1
No. runs	100:10, 200:9, ..., 1000:1	1
Set size	10%:2, 20%:3, ...,90%:1	1
range (max-min)	1:10, 2:9, ..., 9:2, 10:1	1
std	1:10, 2:9, ..., 9:2, 10:1	1
maximum	1:5.5, 2:6.0, ..., 9:9.5, 10:10.0	1
minimum	1:−4.5, 2:−4.0, ..., 9:−0.5, 10:0.0	1
mean	1:5.5, 2:6.0, ..., 9:9.5, 10:10.0	1

Our method assigns scores to the values of eight items of a simulation. Three of them are parameters for its run, and the other five ones are derived from the results of the simulation. The input to our method are simulations with different settings. Our method uses the scores of Table I, sorts the settings according to their scores and gives the setting with the highest score as the best one.

The method gives all items the same weight. The first item is the number of neighbors k. We only evaluated odd values of k from 1 to 19. Our method gives to each of them a score from 1 to 10, respectively. The number of simulations range between 100 and 1000. 100 runs is given a score of 10, 200 times a score of 9 and so on.

The percentage of data for the classification set (also known as non-test set in kNN) is given a score that is small for a small percentage and large for a large one, with the exception that a 90% set is given a score of 1.

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Our method also gives a score to accuracy (correctly predicted percentage of cases). The range between maximum and minimum result values are scaled and assigned an integer value in the range [1, 10] and given a score. A range belonging to scaled-range 1 is given a score of 10 and so on. Our method also uses this criterion for the standard deviation of the results.

The maximum and minimum values of the prediction results are used as correcting or adjusting factors. Both values are also scaled to integer values in the range [1, 10].

The smallest scaled maximum value is given a positive score of 5.5 and the largest one a score of 10.

On the other side, the smallest scaled minimum value of 1 gets a -4.5 score and a 10-value one a score of 0.0 (such that a setting with a high minimum is not affected).

The mean values of the prognosis results are also scaled and given a score similar to the one used for maximum values. All these scores have been adjusted/set experimentally.

We run simulations with 5,130 different settings and tested different scores and items arriving to the values of Table I.

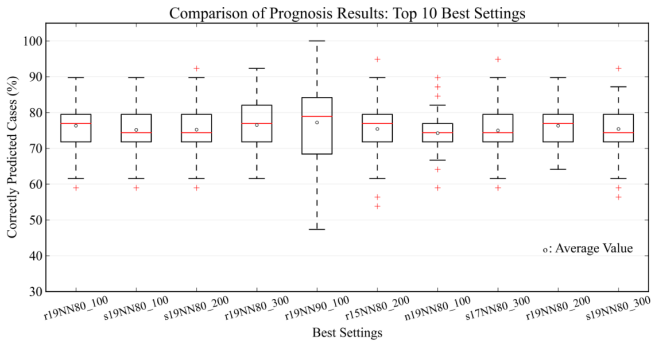

Fig. 7. Top 10 settings of our selection method: Best one on the far left.

In Figure 7 we plotted the top 10 settings obtained with the selection method described above. The first letter in the name of the setting indicates the type of data (r: raw, s: standardized, n: normalized). The number following it is the k value. The number following NN in the name indicates the percentage of data used in the training set, and the last value the number of simulation runs. The lower and upper edges of the boxes in this figure show the 25th and 75th percentile of the data, the horizontal line in them the median, the wiskers extend to the most extreme data points and the data considered outliers are shown individually.

IV. Conclusions

We have shown in this paper some characteristics of the results obtained with the kNN algorithm and proposed a method for evaluating different settings of the kNN algorithm and selecting a good one. Our selection method uses the results obtained with the test data and determines a good setting using scores applied to settings of a run and the result values of it. We evaluated our selection method using the Wisconsin breast cancer prognosis raw data, and its standardized and normalized versions.

Experimental results show that 100 simulation runs are enough to evaluate different settings and that raw and standardized data give better results that an approach using normalized data. Also, classification sets that use 80% of the data and a value of k of 19 have high average, maximum and minimum accuracy values, and a small difference between these last two values. The top setting, according to our selection method, has an average accuracy of 76% a maximum value of 90% and a minimum value of 62%.

In our experiments we used all the nuclei data, but several works use only five of them [1],[9]. Evaluation of our selection method using them is one topic for future research. As noted above there are other distance definitions or measures of similarity [15], the use of them with our selection method is now a topic of an ongoing research work.

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